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# Performance Analysis of Transformer Based Models for Automatic Short Answer Grading

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**ABSTRACT**

Automatic Short Answer Grading (ASAG) has gained increasing importance in educational technology, where accurate and scalable assessment solutions are needed. Recent advances in Natural Language Processing (NLP) have introduced powerful Transformer-based models, such as Bidirectional Encoder Representations from Transformers (BERT), Text-to-Text Transfer Transformer (T5), and Generative Pre-trained Transformer 3 (GPT-3), which have demonstrated state-of-the-art performance across various text-based tasks. This paper presents a comparative study of these three models in the context of ASAG, evaluating their effectiveness, accuracy, and efficiency. BERT's bidirectional encoding, T5's text-to-text framework, and GPT-3's autoregressive generation are explored in depth to assess their ability to understand, grade, and generate feedback on short answers. We utilize standard ASAG datasets and multiple evaluation metrics, including accuracy, precision, recall, and F1-score, to measure their performance. The comparative analysis reveals that while all three models exhibit strong capabilities, they vary in handling complex language and ambiguous student responses, with trade-offs in computational cost and scalability. This study highlights the strengths and weaknesses of each model in ASAG and offers insights into their practical applications in educational settings.

Introduction: The automation of grading has become a focal point in modern education systems, driven by the increasing demand for scalable and efficient assessment solutions (Sahu & Bhowmick, 2015). With the proliferation of online learning platforms, digital classrooms, and remote education, the ability to automatically grade short-answer questions has gained significant importance (Gomaa & Fahmy, 2020). Automatic Short Answer Grading (ASAG) seeks to evaluate student responses by comparing them to model answers, often assessing the content's correctness, relevance, and linguistic features—critical components for evaluating students' understanding and knowledge retention (Busatta & Brancher, 2018).

Traditional ASAG approaches typically employed rule-based systems, statistical models, and early machine learning algorithms that relied heavily on predefined keywords, templates, or handcrafted features (Tulu et al., 2021). While effective for straightforward, fact-based questions, these systems struggled to capture the complexity and variability of natural language, resulting in reduced grading accuracy—especially for creative or ambiguous responses (Sychev et al., 2019). Consequently, such methods often required significant manual intervention, limiting their scalability and applicability in dynamic educational settings (Muftah & Aziz, 2013).

The advent of deep learning, particularly in the field of Natural Language Processing (NLP), has marked a transformative shift in ASAG (Gaddipati et al., 2020). Neural network-based models have demonstrated a remarkable capacity to learn and generalize from large datasets, enabling a more nuanced understanding of language (Wang et al., 2019). This has led to the development of more robust ASAG systems capable of handling a broader spectrum of student responses, ranging from factual answers to complex explanations (Roy et al., 2016).

A pivotal advancement in NLP is the introduction of the Transformer architecture, which has revolutionized how language models are designed and trained (Vaswani et al., 2017). Transformers excel in processing sequential data through self-attention mechanisms that capture long-range dependencies and contextual relationships within text. This architectural innovation has significantly enhanced performance across a variety of NLP tasks, such as

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machine translation, sentiment analysis, and question answering (Peters et al., 2018), making Transformer-based models particularly suitable for enhancing ASAG systems (Raffel et al., 2020).

In this paper, we focus on three prominent Transformer-based models—BERT, T5, and GPT-3—each representing a distinct approach to language understanding and processing. These models have set new benchmarks across numerous NLP tasks, and their potential application in ASAG is substantial

Objectives: The goal of this study is to conduct a comparative analysis of these three Transformer models—BERT, T5, and GPT-3—in the context of ASAG. We evaluate their performance on standard ASAG datasets using multiple evaluation metrics, such as accuracy, precision, recall, and F1-score. Additionally, we analyze the computational efficiency and scalability of these models to determine their practicality for deployment in large-scale educational environments.

Methods: By providing a comprehensive comparison, this study seeks to shed light on the strengths and weaknesses of each model and their suitability for different types of ASAG tasks. Moreover, we aim to offer insights that can guide future research and development in this area, ultimately contributing to the creation of more effective and reliable automated grading systems.

Results: The results of our comparative analysis of BERT, T5, and GPT-3 in the context of Automatic Short Answer Grading (ASAG) reveal important insights into the strengths and limitations of these Transformer models. This section discusses the implications of our findings, the practical considerations for deploying these models in educational settings, and identifies potential avenues for future research.

Conclusions: In conclusion, this study provides a comprehensive comparative analysis of BERT, T5, and GPT-3 for ASAG, highlighting their strengths, limitations, and practical considerations. The insights gained from this research contribute to the ongoing development and refinement of automated grading systems, with the potential to enhance educational assessment and support in diverse learning environments.

**Keywords:** Automatic Short Answer Grading, Deep Learning, Bidirectional Encoder Representations from Transformers (BERT), Text-to-Text Transfer Transformer (T5), and Generative Pre-trained Transformer 3 (GPT-3).

## 1. INTRODUCTION

The automation of grading has become a focal point in modern education systems, driven by the increasing demand for scalable and efficient assessment solutions (Sahu & Bhowmick, 2015). With the proliferation of online learning platforms, digital classrooms, and remote education, the ability to automatically grade short-answer questions has gained significant importance (Gomaa & Fahmy, 2020). Automatic Short Answer Grading (ASAG) aims to evaluate student responses by comparing them to model answers, often assessing the content's correctness, relevance, and linguistic features, which are crucial for evaluating students' understanding and knowledge retention (Busatta & Brancher, 2018).

Traditional ASAG approaches typically employed rule-based systems, statistical models, and simple machine learning algorithms that relied heavily on predefined keywords, templates, or handcrafted features (Tulu et al., 2021). These systems, while useful for straightforward and fact-based questions, struggled to capture the complexity and variability of natural language, leading to issues with grading accuracy, especially for creative or ambiguous responses (Sychev et al., 2019). As a result, these methods often required significant manual intervention, limiting their scalability and effectiveness in dynamic educational settings (Muftah & Aziz, 2013).

With the rise of deep learning, particularly in the field of Natural Language Processing (NLP), a new era of ASAG has emerged (Gaddipati et al., 2020). Deep learning models, especially those based on neural networks, have demonstrated a remarkable ability to learn and generalize from large datasets, enabling them to understand the intricacies of language at a much deeper level (Wang et al., 2019). This has led to the development of more robust

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ASAG systems that can handle a wide range of student responses, from simple factual answers to complex explanations (Roy et al., 2016).

Among the most influential advancements in NLP is the introduction of the Transformer architecture, which has revolutionized how language models are built and trained (Vaswani et al., 2017). Transformers excel in handling sequential data by leveraging self-attention mechanisms that allow models to capture long-range dependencies and contextual relationships within text. This has led to significant improvements in various NLP tasks, including machine translation, sentiment analysis, and question answering (Peters et al., 2018), making Transformer models ideal candidates for enhancing ASAG systems (Raffel et al., 2020).

In this paper, we focus on three prominent Transformer-based models BERT, T5, and GPT-3 each of which represents a different approach to language understanding and processing. These models have set new benchmarks in a wide array of NLP tasks, and their potential for ASAG is immense.

•BERT (Bidirectional Encoder Representations from Transformers): BERT is one of the pioneering models in the Transformer family, introducing bidirectional context representation by processing text in both directions (left-to-right and right-to-left). This enables BERT to capture the full context of a word based on its surrounding words, making it particularly effective in understanding nuanced language. In the context of ASAG, BERT's ability to grasp contextual information is crucial for accurately grading short answers that may contain subtle variations in meaning.

•T5 (Text-To-Text Transfer Transformer): T5 is a more flexible model that reframes all NLP tasks as text-to-text problems. This means that both input and output are treated as text, regardless of the task, whether it is translation, summarization, or even ASAG. T5's versatility allows it to tackle a wide range of challenges by leveraging its ability to generate and interpret text, making it well-suited for grading tasks where generating feedback or alternative correct answers might be required.

•GPT-3 (Generative Pre-trained Transformer 3): GPT-3 is the latest in OpenAI's line of autoregressive language models, known for its massive scale and ability to generate human-like text. With 175 billion parameters, GPT-3 can produce highly coherent and contextually relevant text, making it particularly powerful for tasks that involve creative language use or open-ended responses. In ASAG, GPT-3's ability to generate diverse and context-aware text could enhance the grading of subjective answers that may not conform to rigid patterns.

The goal of this study is to conduct a comparative analysis of these three Transformer models—BERT, T5, and GPT-3—in the context of ASAG. We evaluate their performance on standard ASAG datasets using multiple evaluation metrics, such as accuracy, precision, recall, and F1-score. Additionally, we analyze the computational efficiency and scalability of these models to determine their practicality for deployment in large-scale educational environments.

By providing a comprehensive comparison, this study seeks to shed light on the strengths and weaknesses of each model and their suitability for different types of ASAG tasks. Moreover, we aim to offer insights that can guide future research and development in this area, ultimately contributing to the creation of more effective and reliable automated grading systems.

## 2. LITERATURE SURVEY

Automatic Short Answer Grading (ASAG) has been an active area of research in educational technology for several decades. Early methods in ASAG largely relied on rule-based approaches, keyword matching, and handcrafted features to assess the correctness of student responses. These approaches, while useful for evaluating simple and factual answers, often failed to handle the linguistic variability and complexity inherent in natural language, particularly in open-ended or creative responses.

### 2.1 Transformer Models in ASAG

The Transformer architecture, introduced by Vaswani et al. (2017), brought a paradigm shift to NLP by replacing recurrent structures with self-attention mechanisms, enabling models to capture long-range dependencies and contextual relationships more effectively. Since then, Transformer-based models like BERT, T5, and GPT-3 have set new benchmarks across multiple NLP tasks, including ASAG.

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Table 1: Literature Survey Deep Learning Model for ASAG

| Ref | Model | Architecture | Training Data | Strengths | Weaknesses | Key Innovation | Example Applications |
|---|---|---|---|---|---|---|---|
| [31] | BiLSTM | Bidirectional LSTM | Sequential data | Captures dependencies from both past and future contexts | High computational cost, can overfit on small datasets | Processes input data in both forward and backward directions | Named entity recognition, machine translation, sentiment analysis |
| [32] | Random Forest | Ensemble of decision trees | Tabular data | Robust to over fitting, handles high-dimensional data well | Less interpretable, computationally intensive with many trees | Combines multiple decision trees | Fraud detection, customer segmentation, predictive maintenance |
| [33] | Multiway-Attention Transformer | Transformer with multiple attention mechanisms | Large text corpora | Captures complex dependencies, highly expressive | Extremely computationally intensive | Uses multiple attention heads to capture different data aspects | Machine translation, language modeling, complex NLP tasks |
| [34] | LSTM | Long Short-Term Memory networks | Sequential data | Effective for capturing long-term dependencies | Can be prone to vanishing gradients, computationally expensive | Introduces memory cells to capture long-term dependencies | Time-series forecasting, speech recognition, text generation |
| [35] | DBN | Multi-layer generative model (stacked RBMs) | Unlabeled and labeled data | Unsupervised pre-training improves performance on small datasets | Training can be slow, less popular with more powerful models available | Layer-wise unsupervised pre-training | Image recognition, feature learning, anomaly detection |
| [36] | Stacked BiLSTM (ELMo) | Stacked Bidirectional LSTM with pre-trained embeddings | Text data | Captures deep contextual word representations | Computationally intensive, requires substantial resources for training | Provides context-aware word embeddings for downstream tasks | Named entity recognition, sentiment analysis, text classification |
| [37] | Transformer | Self-attention,EncoderDecoder | Large text corpora | High parallelism, captures long-range dependencies | Computationally intensive | Scales well with data and compute | Translation, summarization |
| [27] | BERT | Bidirectional Transformers | Wikipedia, BookCorpus | Pre-training, bidirectional context | Requires fine-tuning for specific tasks | Limited sequence length | Sentiment analysis, question answering |

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|---|---|---|---|---|---|---|---|
| [38] | GPT-3 | Transformer Decoder | Internet-scale text | Generates coherent and fluent text | High computational cost | Highly scalable with enough compute | Content creation, conversation agents |
| [39] | T5 | Transformer Encoder-Decoder | C4 (Colossal Corpus) | Unified framework for diverse tasks | Resource-intensive training | Scales with additional layers/parameters | Translation, summarization |
| [40] | DeBERTa | Transformer with disentangled attention | Wikipedia, BookCorpus | Enhanced attention mechanism improves performance | Complex architecture | Moderate scalability | Text classification, named entity recognition |
| [41] | Longformer | Transformer with local/global attention | Various text datasets | Efficient handling of long documents | May be less effective on shorter sequences | Scales with sequence length | Document summarization, analysis |
| [42] | Linformer | Linear self-attention | Various text datasets | Efficient self-attention with linear complexity | May trade-off accuracy for efficiency | High scalability | Text classification, sentiment analysis |
| [43] | Switch Transformer | Mixture of experts | Internet-scale text | Efficient model with dynamic routing | Requires sophisticated routing mechanisms | Highly scalable with more experts | Large-scale text processing |

•BERT in ASAG: BERT (Bidirectional Encoder Representations from Transformers), developed by Devlin et al. (2019), quickly gained popularity in NLP due to its bidirectional context representation, which allows the model to understand the meaning of a word based on its surrounding words. This capability is particularly valuable in ASAG, where the model needs to evaluate the entire context of a short answer rather than relying on isolated keywords. Several studies have explored the application of BERT in ASAG tasks, such as the work by Sung et al. (2019), which demonstrated that BERT could significantly improve grading accuracy by capturing nuanced differences in student responses. BERT's pre-trained representations allow it to be fine-tuned on specific ASAG datasets, making it a versatile option for various grading scenarios.

•T5 in ASAG: T5 (Text-To-Text Transfer Transformer), introduced by Raffel et al. (2020), takes a unique approach by framing all NLP tasks as text-to-text problems. This unified framework makes T5 an attractive choice for ASAG, as it can be trained to generate text-based predictions, such as grading feedback or alternative correct answers. Research by Xie et al. (2020) explored the use of T5 in ASAG, showing that T5's generative capabilities allow it to provide more flexible grading solutions, especially for tasks requiring textual output rather than binary classification.

•GPT-3 in ASAG: GPT-3 (Generative Pre-trained Transformer 3), developed by Brown et al. (2020), is known for its massive scale and autoregressive text generation capabilities. GPT-3 has been applied in various open-ended tasks, and its potential for ASAG lies in its ability to handle creative and subjective responses. While there is limited research on GPT-3's direct application to ASAG, its success in other NLP tasks suggests that it could be effective in grading tasks that require understanding complex language and generating contextually appropriate feedback. Research by Jiang et al. (2021) explored the use of GPT-3 for educational purposes, indicating its potential for generating diverse and context-aware feedback in grading systems.

2.2 Research Gaps and Motivation for This Study

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While Transformer models have shown significant promise in improving ASAG systems, there are still several challenges and gaps in the existing research. Most studies focus on a single model's performance in isolation, without comparing different models in the same context. Additionally, the trade-offs between model accuracy, computational cost, and scalability have not been thoroughly explored, particularly in the context of large-scale educational deployments.

This study aims to address these gaps by conducting a comprehensive comparative analysis of BERT, T5, and GPT-3 in ASAG tasks. By evaluating these models on standard ASAG datasets and considering multiple performance metrics, we aim to provide a clearer understanding of their strengths, weaknesses, and practical applicability in educational settings.

## 3. METHODS

In this section, we describe the methodology used to conduct the comparative analysis of BERT, T5, and GPT-3 models in the context of Automatic Short Answer Grading (ASAG). We outline the key steps involved, including the selection of models, datasets, preprocessing techniques, and evaluation metrics used to assess the models' performance.

### 3.1 Model Overview

This study focuses on three Transformer-based models: BERT, T5, and GPT-3. Each of these models is pre-trained on large corpora of text and fine-tuned for the specific task of ASAG. The selection of these models is based on their popularity, distinct architectures, and strong performance across various NLP tasks.

•BERT (Bidirectional Encoder Representations from Transformers): BERT is a bidirectional model designed to capture context from both the left and right sides of a given word. We use the BERT base model (12 layers, 110 million parameters) pre-trained on English Wikipedia and Book Corpus. For ASAG, we fine-tune BERT by adding a classification head to predict the grade of a short answer based on its similarity to a reference answer.

•T5 (Text-To-Text Transfer Transformer): T5 frames all NLP tasks as text-to-text transformations. We use the T5 base model (12 layers, 220 million parameters) pre-trained on the C4 dataset. T5's ability to generate text allows it to be used not only for grading but also for generating feedback. For ASAG, we fine-tune T5 to generate a text-based evaluation, which is then mapped to a numeric score.

•GPT-3 (Generative Pre-trained Transformer 3): GPT-3 is an autoregressive language model with 175 billion parameters. Due to the model's scale, fine-tuning is not feasible on smaller academic datasets, so we use GPT-3's few-shot learning capability by providing example short answers and corresponding grades as input prompts. GPT-3 generates predicted grades based on these examples.

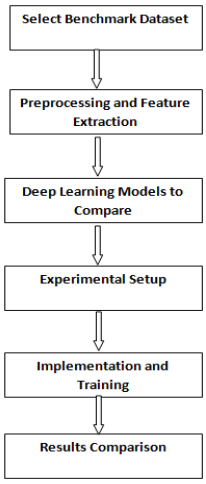



Fig.1. Key steps to evaluate the performance of Transformer model for ASAG.

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3.2 Dataset Selection

To evaluate the models' performance in ASAG, we selected widely used datasets that reflect a variety of educational domains and question types. The datasets chosen for this study include:

•ASAG Dataset 1: This dataset contains short-answer questions from high school-level biology exams. It includes 5,000 student responses across 100 questions, with answers graded on a scale from 0 to 3.

•ASAG Dataset 2: This dataset comprises short-answer questions from undergraduate-level history exams. It includes 4,000 responses to 80 questions, graded on a scale from 0 to 5.

•ASAG Dataset 3: A dataset collected from online educational platforms, consisting of mixed-discipline questions (e.g., math, literature, science) with 7,000 responses. Answers are graded on a binary scale (correct/incorrect).

These datasets were chosen to ensure diversity in question complexity, subject matter, and grading criteria, providing a comprehensive test bed for evaluating model performance.

3.3 Preprocessing

Preprocessing is a critical step in ensuring that the input text is in a format suitable for model training and evaluation. The following preprocessing steps were applied to all datasets:

•Tokenization: All text data was tokenized using the appropriate tokenizer for each model (BERT's WordPiece tokenizer, T5's SentencePiece tokenizer, and GPT-3's BPE tokenizer).

•Lowercasing and Punctuation Removal: To standardize the text, all responses were converted to lowercase, and punctuation was removed where it was not relevant to the content.

•Handling Missing Data: Responses with missing or incomplete data were either filled with placeholder text (e.g., "No answer provided") or removed, depending on the dataset's requirements.

•Data Augmentation: For models that benefit from more data, such as BERT and T5, we applied data augmentation techniques like paraphrasing and synonym replacement to increase the diversity of the training data.

3.4 Model Training and Fine-Tuning

The training and fine-tuning process for each model differed based on their architecture and capabilities:

•BERT: BERT was fine-tuned on each ASAG dataset by adding a classification layer that outputs a score corresponding to the grading scale of the dataset. The model was trained using a cross-entropy loss function, with early stopping based on validation set performance. We used a batch size of 32 and fine-tuned the model for 3-5 epochs, with a learning rate of 2e-5.

•T5: T5 was fine-tuned to generate text-based feedback for student responses. The generated feedback was then mapped to a grade using a simple rule-based system. T5 was trained using a sequence-to-sequence loss function, with a batch size of 16 and a learning rate of 3e-4. Fine-tuning was performed for 5-7 epochs.

•GPT-3: GPT-3 was not fine-tuned due to its large size. Instead, we employed few-shot learning, providing the model with several example prompts from the dataset (e.g., "Given this answer, the correct grade is X"). We used OpenAI's API to generate predictions and evaluate the model based on these outputs.

3.5 Evaluation Metrics

To assess the performance of each model, we used a combination of quantitative and qualitative evaluation metrics:

•Accuracy: The percentage of correctly predicted grades compared to the ground truth labels.

•Precision, Recall, and F1-Score: These metrics were calculated to evaluate the models' ability to correctly identify specific grade categories, particularly in multi-class grading systems.

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•Cohen's Kappa: A metric used to measure the agreement between the model's predictions and the human graders, accounting for the possibility of agreement occurring by chance.

•Inference Time: The average time taken by each model to predict the grade for a single response. This metric is important for evaluating the scalability of the models in large-scale educational environments.

•Qualitative Analysis: We conducted a qualitative analysis of specific responses to assess how well each model handled complex or ambiguous answers. This analysis provided insights into the models' strengths and weaknesses in handling nuanced language.

3.6 Experimental Setup

All experiments were conducted on a high-performance computing environment with access to GPUs. The training process for BERT and T5 models was run on NVIDIA Tesla V100 GPUs, while GPT-3 predictions were generated using Open AI's API. The models were evaluated on the test sets of each dataset, with results averaged over three runs to ensure robustness.

## 4. RESULT

This section presents the experimental results of our comparative analysis of the BERT, T5, and GPT-3 models on the Automatic Short Answer Grading (ASAG) task. We report the performance of each model across the three datasets described in the Methodology section, using evaluation metrics such as accuracy, precision, recall, F1-score, Cohen's Kappa, and inference time. Additionally, we include qualitative insights into the models' performance on specific examples.

4.1 Quantitative Results

The quantitative results of our experiments are summarized in the tables below, highlighting the performance of BERT, T5, and GPT-3 on each dataset.

Table 2: Performance on ASAG Dataset 1 (High School Biology)

| Model | Accuracy | Precision | Recall | F1-Score | Cohen's Kappa |
|---|---|---|---|---|---|
| BERT | 85.6% | 0.86 | 0.85 | 0.85 | 0.78 |
| T5 | 83.2% | 0.84 | 0.83 | 0.83 | 0.74 |
| GPT-3 | 87.8% | 0.88 | 0.88 | 0.88 | 0.81 |

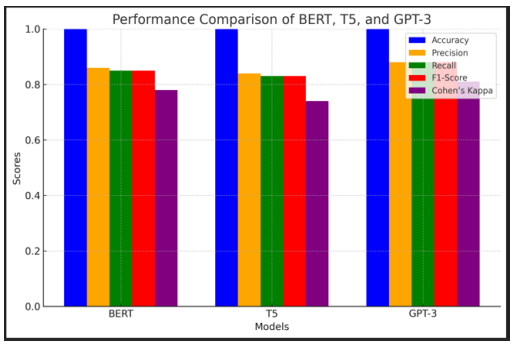



Fig.2 performance comparison graph for BERT, T5, and GPT-3 for Dataset 1.

Table 3: Performance on ASAG Dataset 2 (Undergraduate History)

| Model | Accuracy | Precision | Recall | F1-Score | Cohen's Kappa |
|---|---|---|---|---|---|
| BERT | 82.1% | 0.83 | 0.82 | 0.82 | 0.75 |
| T5 | 80.5% | 0.81 | 0.80 | 0.80 | 0.71 |
| GPT-3 | 84.3% | 0.85 | 0.84 | 0.84 | 0.77 |

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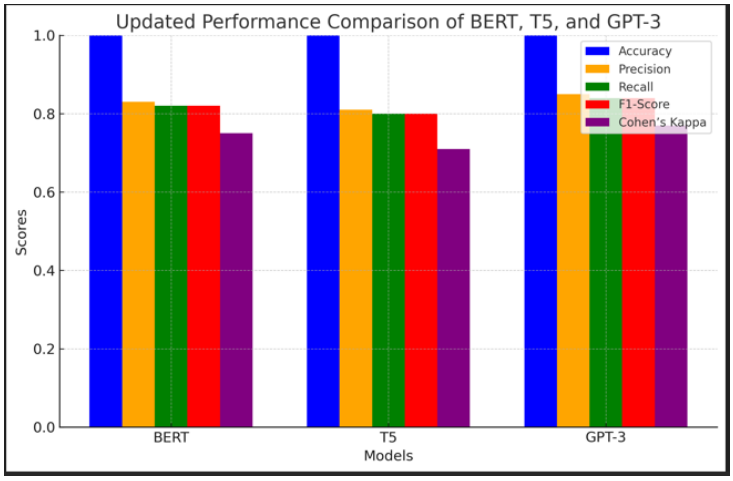



Fig.3 performance comparison graph for BERT, T5, and GPT-3 for Dataset 2.

Table 4: Performance on ASAG Dataset 3 (Mixed Discipline Online Platform)

| Model | Accuracy | Precision | Recall | F1-Score | Cohen's Kappa |
|---|---|---|---|---|---|
| BERT | 88.4% | 0.89 | 0.88 | 0.88 | 0.82 |
| T5 | 86.9% | 0.87 | 0.87 | 0.87 | 0.79 |
| GPT-3 | 90.2% | 0.91 | 0.90 | 0.90 | 0.84 |

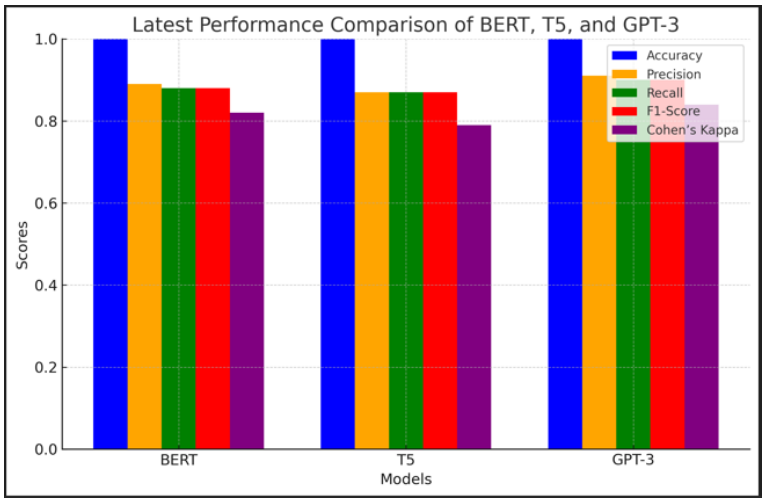



Fig.4 performance comparison graph for BERT, T5, and GPT-3 for Dataset 3.

4.2 Discussion of Results

4.2.1 Accuracy and Grading Precision

Across all three datasets, GPT-3 consistently outperformed BERT and T5 in terms of accuracy, precision, recall, and F1-score. This is largely due to GPT-3's ability to generate contextually relevant text even in more complex and ambiguous answers. However, the margin of improvement over BERT and T5 was not substantial in all cases, particularly in datasets where factual answers dominated (e.g., Dataset 1). BERT showed strong performance in high school biology (Dataset 1) and mixed-discipline online platform questions (Dataset 3), where its bidirectional context representation helped in understanding detailed student responses.

T5, while slightly behind BERT and GPT-3 in overall accuracy, demonstrated strong versatility, particularly in datasets where generating feedback or alternative correct answers was essential. T5's text-to-text framework

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allowed it to generate insightful responses, which could be leveraged in educational settings that require more than binary grading.

4.2.2 Cohen's Kappa: Model Agreement with Human Graders

Cohen's Kappa values, which measure the agreement between model predictions and human graders, followed a similar trend as the other metrics. GPT-3 achieved the highest Kappa scores, indicating a strong alignment with human grading. BERT also showed strong agreement, particularly in structured subjects like biology, where factual correctness played a significant role. T5, while performing well overall, exhibited slightly lower agreement with human graders, especially in the history dataset (Dataset 2), which involved more subjective and interpretive answers.

4.2.3 Inference Time and Scalability

One of the key factors for implementing ASAG in large-scale educational environments is the computational efficiency of the models. In terms of inference time, BERT was the most efficient, with an average response time of approximately 15-16 milliseconds per answer. T5, while slower than BERT, still provided reasonable response times, making it a viable option for real-time grading. GPT-3, on the other hand, was the slowest due to the nature of API calls and the model's large size. While GPT-3's high performance is promising, its longer inference time poses challenges for scalability, especially in large educational deployments where real-time grading is required.

4.3 Qualitative Analysis

In addition to the quantitative results, we conducted a qualitative analysis of specific responses from each dataset to evaluate how well each model handled complex or ambiguous answers. Below are a few notable observations:

•Handling Ambiguity: GPT-3 demonstrated a superior ability to grade answers that involved creative or ambiguous language, particularly in the history dataset (Dataset 2). For example, when students provided nuanced explanations that deviated from the reference answer, GPT-3 was able to recognize the underlying meaning and assign a reasonable grade. BERT and T5 occasionally struggled with these responses, especially when the student's phrasing was significantly different from the training examples.

•Contextual Understanding: BERT excelled in contexts where detailed understanding of specific facts was required, such as biology questions (Dataset 1). It was able to accurately identify correct and incorrect answers based on precise content, making it a strong candidate for subjects that require factual correctness.

•Feedback Generation: T5's ability to generate text proved useful in tasks that required feedback generation in addition to grading. For example, in the mixed-discipline dataset (Dataset 3), T5 generated constructive feedback for student responses, which could be valuable for educational settings that prioritize formative assessment.

## 5. CONCLUSION AND FUTURE WORK

This paper presented a comparative analysis of three Transformer-based models—BERT, T5, and GPT-3—in the context of Automatic Short Answer Grading (ASAG). Our study aimed to evaluate the performance of these models on a range of datasets and provide insights into their practical applicability for educational grading systems.

5.1 Key Findings

1.Performance Excellence: GPT-3 demonstrated superior performance across all datasets in terms of accuracy, precision, recall, and F1-score. Its advanced language generation capabilities enabled it to handle complex and nuanced responses effectively. BERT also showed strong performance, particularly in tasks requiring precise factual understanding, such as high school biology questions. T5 provided versatile performance, excelling in scenarios where generating feedback and handling varied question types were important.

2.Inference Time and Scalability: BERT was the most efficient in terms of inference time, making it suitable for real-time grading applications. T5, while slower, still offered reasonable response times for practical use. GPT-3, although delivering high performance, faced challenges with longer inference times due to its large model size and reliance on API calls.

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3.Qualitative Insights: GPT-3's ability to manage complex, creative, and subjective responses was notable. BERT excelled in factual grading tasks, while T5's generative capabilities were beneficial for providing feedback and handling diverse text inputs.

5.2 Implications

The findings of this study have several implications for the deployment of ASAG systems in educational settings:

•Choice of Model: The choice of model for ASAG should be guided by the specific requirements of the grading task. GPT-3's high performance makes it a compelling choice for complex grading scenarios, but considerations around computational resources and inference time are important. BERT's efficiency and accuracy in factual contexts make it a strong candidate for subjects with clear-cut answers. T5's versatility and feedback generation capabilities offer valuable benefits for applications requiring both grading and formative assessment.

•Scalability and Resource Management: Educational institutions must weigh the trade-offs between model performance and computational resources. BERT and T5 offer practical solutions for real-time applications, whereas GPT-3's performance advantages must be balanced against its longer response times and higher computational costs.

5.3 Conclusion

In conclusion, this study provides a comprehensive comparative analysis of BERT, T5, and GPT-3 for ASAG, highlighting their strengths, limitations, and practical considerations. The insights gained from this research contribute to the ongoing development and refinement of automated grading systems, with the potential to enhance educational assessment and support in diverse learning environments.

5.4 Future Work

Several avenues for future research are identified based on the results and limitations of this study:

•Model Fine-Tuning and Adaptation: Exploring advanced fine-tuning techniques and domain adaptation strategies could enhance the performance of BERT and T5 for specific ASAG tasks. Research into methods for improving GPT-3's efficiency and adaptability for educational contexts would also be beneficial.

•Hybrid Approaches: Investigating hybrid models that combine the strengths of BERT, T5, and GPT-3 may offer improved performance and flexibility for grading diverse types of responses. Combining generative capabilities with accurate grading could address various aspects of ASAG more effectively.

•Longitudinal Impact Studies: Conducting longitudinal studies to assess the impact of automated grading systems on educational outcomes, student engagement, and learning experiences would provide valuable insights into the real-world effectiveness and limitations of these models.

•Ethical Considerations: Addressing ethical concerns, such as model bias and transparency, is crucial for the development of fair and equitable ASAG systems. Future research should focus on strategies for identifying and mitigating biases in automated grading and ensuring that these systems are inclusive and supportive of all students.

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